

Responsibility-Carrying State: Auditable Sufficiency, Reopen Contracts, and Safe State Reuse

ORION-P13

21 August 2026

Abstract

A compact state is not simply “sufficient” or “insufficient”; sufficiency is relative to the downstream responsibility. We formalize this exact conditional notion and an RCS interface whose declared support controls reuse and reopening. The paper retains an earlier frozen negative and an exactly reproducible P13A benchmark. That benchmark reports zero RCS unsafe reuse, but a later hostile audit changes its authority: RCS reuses exactly when the certificate says **supported**, while the harm counter calls reuse unsafe exactly when the same certificate says **not supported**. Across 3,840 enumerated points, certificate corruption moves the RCS action on 2,304 and the published harm on zero, leaving zero reachable harm opportunities. The outcome-entailment adjudication therefore withholds P13A empirical safety-cost authority. A prospectively frozen P13B then grades against locally authored certificate-independent gold. Across a 30-case state-task panel, each of four corruption worlds has 30 live mutation opportunities; authenticated RCS makes zero unsafe reuses in every world and costs 0.6111 times always raw on valid certificates. Current authority is the authenticated-certificate controlled finite-world result, not external validation.

1 Introduction

Compression, abstraction and state reuse are normally evaluated against a named objective. Problems appear when compact state is reused after the downstream responsibility changes. A summary sufficient for a factual question may omit provenance needed to authorize a scientific claim. A proof-state abstraction useful for next-tactic prediction may omit dependencies needed to diagnose or repair a failed proof. A control-state abstraction adequate for one policy may collapse distinctions needed for intervention or counterfactual analysis.

This is not merely an uncertainty problem. A system can be highly confident while state omits a variable that becomes decisive under a new responsibility. Nor is it merely provenance: knowing exactly where state came from does not establish what that state can safely support.

P13 asks:

Can a compact state carry a machine-checkable responsibility contract that says what it supports, what it omits, what changes invalidate reuse, and whether richer state can be reopened—without forcing every decision to reload raw evidence?

The paper contributes responsibility-relative sufficiency; a responsibility-carrying state interface; permanent negative-result analysis; and an outcome-contingency adjudication that separates the exact supported core from descriptive P13A measurements whose empirical safety-cost authority is withheld.

2 Donor boundary and responsibility-relative sufficiency

Statistical sufficiency, state abstraction, bisimulation, predictive-state representations and causal abstractions already establish that different tasks can require different retained distinctions. Selective prediction and uncertainty gating use confidence to abstain. Provenance/evidence tracing binds artifacts to origin. Proof-carrying systems attach verifiable certificates. Memory-staleness work asks when stored information is no longer valid. P13 claims none of these primitives.

The residual is **responsibility-scoped certified reuse with explicit reopen semantics and exact responsibility-support conditions**. Empirical safety-cost superiority remains unestablished because P13A’s published harm endpoint was self-scored and had no reachable opportunities.

Let raw world X induce correct output $g_rho(X)$ for responsibility rho , and let compact representation be $Z=T(X)$.

2.1 Exact responsibility sufficiency

Z is sufficient for responsibility rho over world set Ω if there exists h_rho such that $g_rho(x)=h_rho(T(x))$ for every x in Ω . Equivalently, every equivalence class induced by T must be homogeneous under g_rho .

2.2 Responsibility-shift witness

A pair (x, x') witnesses insufficiency after $rho_L \rightarrow rho_H$ when $T(x)=T(x')$, the lower-responsibility outputs agree, and the higher-responsibility outputs differ. No learner is needed to establish such a witness; it is a property of the abstraction and responsibility.

For empirical systems, operational sufficiency debt is the verified higher-responsibility performance gap between richer and compact state, conditional on a prospectively frozen lower-responsibility equivalence/noninferiority requirement. It is a benchmark quantity, not a universal information measure.

3 Permanent negative history

The historical experiment constructed independent binary latent variables (x, m, r) and responsibilities $PREDICT=x$, $DECIDE=x$, $INTERVENE=x*m$, $VERIFY=x*m$, and $REPAIR=r$. Representations were $Z1=(x)$, $Z2=(x, m)$ and $Z3=(x, m, r)$.

Exact enumeration produced the intended responsibility ladder: all representations were perfect for prediction/decision; $Z1$ was exactly 0.5 on intervene/verify while $Z2/Z3$ were 1.0; $Z2$ was 0.5 on repair while $Z3$ was 1.0. Exact upward debts were therefore +0.50.

However, the frozen protocol also required the maximum deviation among 100 finite-sample sanity replicates of $n=1024$ to be ≤ 0.05 . The observed maximum was 0.0556640625 at replicate 92. The combined terminal remains permanently `P14_CONTROLLED_SUFFICIENCY_DEBT_GATE_NOT_MET`.

3.1 Root cause

The harness grouped a finite sample by compact state, selected the majority target value from that same sample and credited that majority on the same observations, then maximized deviation over 100 replicates. The statistic combines within-sample majority optimism with an extreme-value operation. The correct response is not to change 0.05 after observing 0.0556640625. The

old terminal stays negative. The successor changes the estimand: exact support is evaluated by exhaustive equivalence classes, while efficacy is evaluated on a fresh safety–cost benchmark.

4 ResponsibilityCarryingState contract

An RCS contains compact state plus a fail-closed contract binding raw/source evidence identity; compiler/transform identity and version; exact supported responsibility set; independent witness/certificate identity; intentionally omitted coordinates/information classes; required-same context coordinates; reopen-on-change coordinates; raw recovery/reconstruction availability and freshness; recovery/reopen cost; resource envelope; evaluator identity; authority owner; and an explicit declaration that the object grants no scientific/novelty self-authority.

At reuse time the contract returns:

- `USE_COMPILED` when requested responsibility is supported and all bound conditions hold;
- `REOPEN_REQUIRED` when support does not hold but richer state can be recovered;
- `CANNOT_CHECK` when support is absent and recovery/verification cannot establish a safe route.

`CANNOT_CHECK` is a correct scientific disposition, not an error to hide.

5 Independent protected benchmark

The successor retains the interpretable world but evaluates support exactly: Z1 supports `{PREDICT, DECIDE}`; Z2 supports `{PREDICT, DECIDE, INTERVENE, VERIFY}`; and Z3 supports all five responsibilities. The exact equivalence-class matrix is checked directly and matches the registered support sets.

Protected seed: 2026082113. Families: **24**. Episodes per family: **512**. Total protected episodes: **12,288**.

Family probabilities for hidden variables `m` and `r` vary independently over `[0.65,0.95]`. Compact state is Z1 or Z2 with equal probability; requested responsibility is uniform over five tasks. Raw recovery is independently available with probability `0.95`. When compact state lacks a needed coordinate, an unqualified decoder uses the family MAP value, deliberately creating cases that can be high-confidence yet structurally unsupported.

5.1 Baselines

1. `UNQUALIFIED`: always reuse compact state.
2. `CONFIDENCE_ONLY`: reuse if estimated task accuracy is at least `0.80`; otherwise reopen when possible.
3. `PROVENANCE_ONLY`: valid lineage is present, so provenance alone permits reuse.
4. `RCS`: reuse only when exact responsibility support holds; otherwise reopen if raw is recoverable, else `CANNOT_CHECK`.
5. `ALWAYS_RAW`: reopen raw state whenever possible.

Fixed resource units are `REUSE=1, REOPEN=6, CANNOT_CHECK=0.5`.

Baselines 1 and 3 are the same policy on this corpus, and the list is four distinct policies rather than five. Every episode supplies valid lineage by construction, so the provenance check never refuses and `PROVENANCE_ONLY` reduces to "always reuse" — which is `UNQUALIFIED`. Their measured rates are identical on every metric reported: unsafe reuse `0.3961588541666667`, verified correctness

0.9248046875, mean cost 1.0. This follows from the construction above rather than from anything that happened during the run, and the results report the two together for that reason. It has one consequence worth stating plainly: this benchmark does not test whether a provenance check helps, because the check is never exercised. Establishing that would need episodes with absent or broken lineage, which this corpus does not contain.

6 Results

The historical runner printed its preregistered safety-cost terminal. The later outcome-entailment adjudication withholds that empirical claim authority. The current bounded authority comes from P13B’s authenticated-certificate controlled finite-world result.

arm	unsafe reuse	verified correctness	unnecessary reopen	mean cost
RCS	0.0000	0.9807	0.0000	2.8747
confidence only	0.2156	0.9657	lower than RCS	1.8582
provenance only	0.3962	0.9248	0	1.0000
unqualified compact	0.3962	0.9248	0	1.0000
always raw	0.0000	0.9513	0.5744	5.7319

RCS emits `CANNOT_CHECK` for all **237** certificate-declared unsupported/nonrecoverable cases and no other protected case. These action, rate, cost and replay numbers are descriptive historical facts. They do not establish elimination of unsafe reuse because the harm endpoint is self-scored.

6.1 Outcome-contingency adjudication

On the exact 3,840-point audit space, a registered certificate corruption moves the RCS action on 2,304 points but moves the published unsafe-reuse endpoint on zero. The endpoint has zero opportunities and returns `CANNOT_CHECK`. A control that grades the same reuse decisions against independently defined gold support has 1,536 live opportunities and passes, establishing that the missing denominator is specific to P13A’s scorer rather than inevitable.

6.2 Why provenance fails

Every compact state has valid lineage. `PROVENANCE_ONLY` and `UNQUALIFIED` are identical policies and identical result rows here, so path or label multiplicity cannot create two independent comparisons.

6.3 Why always raw is not the answer

Always reopening has the reported historical P13A cost/action profile. P13A cannot establish the safety coordinate because its scorer was not independently gradable.

6.4 Prospectively frozen P13B

P13B defines gold support from task requirements and state variables without reading the certificate. The complete panel contains six state forms and five tasks. Omitted, overbroad, forged and stale-epoch worlds each have 30 live mutation opportunities before scoring. Authenticated RCS rejects every mutated certificate and makes zero gold-scored unsafe reuses in every world; unverified RCS has 0, 5, 5 and 16 unsafe reuses in the omitted, overbroad, forged and stale worlds. On valid

certificates, authenticated RCS is exactly correct and costs 0.6111 times always raw. Two fresh subprocess payloads are byte-identical, supporting the registered controlled finite-world claim.

7 Certificate transport, related work and reproducibility

Responsibility support is conditional on evidence identity, transform version, required context, witness identity and resource envelope. A semantic change can require preserve, reopen, revoke or an explicit indeterminate disposition. The RCS object does not certify its own scientific authority. An evaluator can establish operational support; that role grants no novelty, publication or deployment authority.

Recent proof-carrying agent-action work attaches model-agnostic certificates to actions and run-time governance. Provenance traces evidence and execution. Memory-staleness systems detect that stored state is no longer valid. These donors make it insufficient to claim simply that “state should carry a certificate.” P13’s discriminator is the **responsibility key plus reopen semantics**: a state may be current, well-provenanced and high-confidence yet insufficient for another downstream responsibility.

The support matrix is deterministic and exact. The historical negative is reported exactly as frozen and is not reanalysed into a positive result. P13B is a complete registered 30-case finite panel, not a population sample, and reports exact counts rather than a post-hoc inferential interval. Each corruption world must have a nonzero opportunity denominator before zero violations can be interpreted. A real-system extension should define task/domain as the unit of generalization and use paired matched comparisons with family/domain-block uncertainty.

8 Limitations and conclusion

Responsibilities are discrete and known in the controlled world; real systems may have ambiguous or compositional responsibilities. Exact sufficiency is strong; approximate support needs frozen tolerances and calibrated witnesses. The contract is only as reliable as external witness and recovery metadata. Resource costs are controlled units, not measured real latency/tokens/IO. High-risk domains may rationally choose always-raw. The paper has not yet demonstrated verifier-backed Lean repair/diagnosis or blinded scientific-workflow responsibility shifts, and certificate transport/revocation under real semantic version changes remains an external validation gate.

Sufficiency is a contract over future responsibility, not an intrinsic property of compact representation. P13 makes that boundary explicit and preserves its historical failures. P13A’s self-scored harm endpoint remains non-authoritative. P13B shows that the authenticated policy rejects the four registered corruptions without unsafe reuse in its locally authored finite panel while retaining lower valid-panel cost than always raw. Confidence and provenance remain useful signals, but neither specifies what distinctions state supports.

A responsibility/recovery contract can state the exact conditions under which compact-state reuse is authorized; external witness correctness and real-system safety remain separate validation questions.

9 References

- Classical statistical sufficiency, state-abstraction, bisimulation, predictive-state and causal-abstraction literatures are donor-owned foundations.

- Wang, Z. *Proof-Carrying Agent Actions: Model-Agnostic Runtime Governance for Heterogeneous Agent Systems*. arXiv:2606.04104, 2026.
- Chao, H., Bai, Y., Sheng, R., Li, T. & Sun, Y. *STALE: Can LLM Agents Know When Their Memories Are No Longer Valid?* arXiv:2605.06527, 2026.
- Wang, Y. et al. *From Agent Traces to Trust: Evidence Tracing and Execution Provenance in LLM Agents*. arXiv:2606.04990, 2026.
- Doyle, J. *A Truth Maintenance System*. Artificial Intelligence 12(3):231–272, 1979.
- de Kleer, J. *An Assumption-Based TMS*. Artificial Intelligence 28(2), 1986.